A Hybrid Offline-Online Routing Framework for Battery-Constrained Drone Delivery in Wireless Rechargeable Sensor Networks

CS5100 Foundations of Artificial Intelligence — Capstone Final Report
Northeastern University, Khoury College of Computer Sciences

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Submission Links (Public)
Reproduction target: Watkins & Dayan (1992), *Q-learning*, Machine Learning 8:279–292.
Code (GitHub, public): https://github.com/rajayadidiah/GA-DQL
Demo video: https://youtu.be/-O3yH9AKT7g

Abstract. Wireless Rechargeable Sensor Networks (WRSNs) enable persistent environmental monitoring, precision agriculture, and industrial IoT. A central challenge is scheduling a battery-constrained drone to visit energy-depleted sensor nodes before they die, while minimising travel distance, waiting time, and node mortality. In prior B.Tech FYP work (NIT Trichy, 2022) we developed a Genetic Algorithm (GA) charging scheme that outperformed FCFS baselines, but assumed a deterministic environment. This report extends that framework with a **Hybrid Offline-Online Routing Architecture**. Layer 1 retains the GA for nominal path planning with three improvements: tournament selection, adaptive mutation rate decay, and a 2-opt local search. Layer 2 adds a risk-aware, event-triggered Q-Learning recovery agent (reproducing Watkins & Dayan, 1992) that monitors drone state-of-charge (SOC) and selects among local repair, charger diversion, or full recharge. A Double Q-Learning variant reduces overestimation bias; a three-tier severity model (critical/moderate/mild) shapes the reward. Across five test cases (25–100 nodes), the Hybrid GA+Double Q-Learning framework reduces unsafe SOC events by 85–98% vs GA-Only while maintaining or improving AWT. Multi-seed analysis (5 seeds) confirms robustness.

Keywords: WRSN, Genetic Algorithm, Q-Learning, Double Q-Learning, Drone Routing, Reinforcement Learning, Hybrid Optimisation.

1 Introduction

1.1 Background and Motivation

A Wireless Rechargeable Sensor Network (WRSN) consists of battery-powered sensor nodes distributed over a monitoring region that sense physical phenomena and relay data to a base station [5]. The network's operational lifetime is constrained by energy availability. Wireless Energy Transfer technology offers a solution: Mobile Chargers (MCs)—ground vehicles, drones, or robotic platforms—travel among nodes and wirelessly replenish their batteries [7]. The key challenge is determining the charging schedule that optimises:

- **AWT**: nodes waiting too long may die before being charged.
- **Survival Rate**: fraction of nodes alive through the cycle.
- **MC Efficiency**: energy spent charging vs travel overhead.
- **Priorities**: nodes with higher consumption / lower residual energy served first.

This is a TSP-with-time-windows variant, NP-hard [8]; metaheuristics (GA, PSO, ACO) are the practical tools.

1.2 Prior Work: B.Tech FYP at NIT Trichy

In our B.Tech FYP (2022), supervised by Dr. M. Sai Krishna at NIT Trichy, we developed a GA-based Adaptive Charging Scheduling Algorithm (GA-ACSA) [17] with a multi-objective fitness $f(x) = t_{ot} + t_{total} + D + f_1/2 + f_2/2$, roulette-wheel selection with elitism, OX crossover, and swap mutation. Evaluated on five test cases (25–100 nodes), it achieved 30–40% lower AWT, 7–15% higher survival, and 5–18% higher MC efficiency vs FCFS. However, the FYP assumed a **deterministic environment**—a drone executing a pre-computed route will encounter stochastic disruptions that invalidate the plan.

1.3 Limitations of the FYP Approach

(1) No disruption handling (wind, charger blocks, urgent reroutes, battery spikes). (2) Roulette-wheel selection with inconsistent pressure. (3) Fixed mutation rate. (4) No local search refinement. (5) No mechanism for online replanning.

1.4 Reproduction Target and Contributions

The RL component reproduces tabular Q-learning (Watkins & Dayan, 1992) [1] extended with Double Q-Learning [2]. Contributions:

1. **Layer 1 Enhanced GA:** tournament selection, adaptive mutation, 2-opt.
2. **Layer 2 Q-Learning Recovery:** event-triggered agent with three-tier severity.

3. **Double Q-Learning:** reduces maximisation bias.
4. **Learning rate decay:** stabilises Q-table convergence.
5. **Comprehensive evaluation:** 5 test cases, 5 algorithms, 10+ plots, multi-seed analysis.

2 Paper Overview and Literature Review

2.1 Source Paper: Watkins & Dayan (1992)

Watkins and Dayan [1] prove that the one-step off-policy TD update $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$ converges to Q^* with probability one, provided: (i) every state–action pair is visited infinitely often, (ii) α satisfies Robbins–Monro ($\sum_t \alpha_t = \infty$, $\sum_t \alpha_t^2 < \infty$), (iii) finite tabular state space, (iv) stationary MDP. Our Layer-2 agent implements this update with decaying linear α and ε schedules inspired by the Robbins–Monro stability conditions (we use a finite linear decay over 500 episodes, which approximates but does not formally satisfy the infinite-time step-size sum conditions). We stress-test the guarantee by deploying it in a non-stationary MDP with 57% per-step disruption.

2.2 Charging Scheduling in WRSNs

Tomar et al. [5] use fuzzy on-demand charging; Nguyen et al. [16] propose a two-phase scheduler. Xu et al. [8] and Huong et al. [9] apply GAs to multiple charging cars. Feng et al. [12] propose Newborn PSO; Gharaei et al. [7] optimise charger circuitry. Soni and Shrivastava [10], Cao et al. [11], and Wei et al. [6] apply RL. Dong et al. [13] introduce TSCCS; Tian et al. [15] combine charging with node sleep; Lu et al. [14] propose joint routing-and-charging.

2.3 Algorithmic Positioning

We deliberately select *tabular* off-policy Q-learning for the recovery agent because the state abstraction is small ($|S| = 550$) and the 1992 convergence proof is directly testable. Our choice contrasts with neural TD methods such as TD-Gammon [4] (on-policy TD(λ) with a feed-forward approximator) and modern DQN-family approaches. Moving to function approximation would trade the tabular guarantee for scalability—left as future work.

2.4 Literature Gap

(1) No hybrid offline-online framework combining precomputed optimal routes with a learned recovery policy. (2) No graduated disruption response. (3) Most GA-based WRSN works use basic roulette-wheel and fixed parameters.

3 Reproduction Objective

Central question. *Does the 1992 Q-learning convergence claim survive when the tabular MDP is embedded in a realistic, non-stationary routing environment with a 57% per-step disruption rate, and if not, what minimal extensions make it safely deployable?*

Reproduction scope (what we reproduce as-is). (i) The one-step off-policy update Equation (4); (ii) ε-greedy behaviour policy decoupled from the greedy target policy; (iii) a decaying linear α schedule inspired by Robbins–Monro stability conditions; (iv) finite tabular $|S|$=550, $|A|$=3 representation.

Outside original scope (flagged extensions). (a) Double Q-Learning [2] to diagnose overestimation bias; (b) a GA-derived nominal route used to seed the behaviour policy; (c) three-tier risk-aware reward shaping. Each is presented as an explicit design choice, not as a claim of the 1992 paper.

Success criteria. A reproduction is "successful" if (1) the learning curve monotonically improves over ≥ 400 episodes across 5 seeds, (2) the learned greedy policy beats a random-policy baseline on AWT, and (3) the trend holds across all 5 test cases.

4 Problem Formulation

4.1 Network Model

A WRSN on a 200×200 grid: $N = \{n_1, \ldots, n_K\}$ sensor nodes (coords, ECR $\in [1, 10]$, residual energy RE $\in [0, 100]$, priority DL $\in [0, 10]$), one base station (centre), a single drone (initial SOC 80%, $\text{SOC}_{\max} = 100\%$, speed 5 m/s), and five charger stations (queue capacity 3).

4.2 Charging Requests and Objective

Node n_i requests charging when $\text{re}_i < E_{\text{thresh}} = 50$. $R = \{n_i : \text{re}_i < 50\}$.

$$f(\pi) = t_{ot}(\pi) + t_{total}(\pi) + D(\pi) + \frac{f_1(\pi)}{2} + \frac{f_2(\pi)}{2} \quad (1)$$

with t_{ot} overtime, t_{total} total charge duration, D travel distance, f_1 priority-weighted waiting, f_2 fairness deviation. Goal: $\pi^* = \arg\min_\pi f(\pi)$.

4.3 Stochastic Disruption Model

Table 1: Stochastic disruption parameters.

Type	Prob.	SOC Impact
Wind drain	0.20	-5 to -18%
Charger block	0.30	indirect
Urgent request	0.15	$-0.04d$
Battery spike	0.10	-8 to -20%

4.4 Three-Tier Severity

Table 2: Three-tier disruption severity.

Tier	Condition
Critical	SOC $< 25\%$ or projected $< 15\%$
Moderate	$25 \leq$ SOC $< 42\%$ or (blocked & SOC < 55)
Mild	blocked & SOC $\geq 55\%$

5 Methodology

5.1 Layer 1: Enhanced GA

Each chromosome is a permutation of requesting nodes. We replace roulette-wheel with **tournament selection** ($k = 5$) plus elitism. Mutation rate follows an adaptive schedule:

$$p_m(g) = \max(p_{\min},\ p_{\text{base}}(1 - 0.7g/G)) \tag{2}$$

with $p_{\text{base}} = 0.01$, $p_{\min} = 0.002$. After convergence, **2-opt local search** (window 15, max 50 iters) refines the route. OX and Swap Mutation are retained.

5.2 Layer 2: Q-Learning (Watkins & Dayan, 1992)

State. $s = (\text{soc_bucket}, \text{charger_idx}, \text{pending})$, $|S| = 10 \times 5 \times 11 = 550$.

Actions. a=0 local repair; a=1 diversion (+70% SOC); a=2 safety-first full recharge.

Reward. The shaped reward couples the three severity tiers with action choice. Let $\mathbb{1}_v = \mathbb{1}[\text{SOC violation}]$. For the *critical* tier: $r = -150 - 200\mathbb{1}_v$ if a=0, else $3\Delta\text{SOC} + 30\mathbb{1}[\text{SOC}>60] - 50\mathbb{1}_v$. For *moderate*: $-40 - 100\mathbb{1}_v$ if a=0, else $2\Delta\text{SOC} + 20\mathbb{1}[\text{SOC}>60] - 30\mathbb{1}_v$. For *mild*: $15 - 80\mathbb{1}_v$ if a=0, else $5 - 0.02d_c$. We refer to this compound specification as Equation (3):

$$r(s,a) = r_{\text{tier}(s),\,a} \quad \text{(six-case piecewise above).} \tag{3}$$

Q-update (1992).

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)], \tag{4}$$

$\gamma = 0.95$.

Learning rate decay. α decays linearly $0.15 \to 0.01$ over $N_{\text{train}} = 500$ episodes. This finite linear decay is inspired by the Robbins–Monro stability intuition (large early steps, small late steps) but does not formally satisfy its infinite-horizon conditions.

Double Q-Learning.

$$Q_1 \leftarrow Q_1 + \alpha[r + \gamma Q_2(s', \arg\max_{a'} Q_1(s',a')) - Q_1]. \tag{5}$$

One of Q_1, Q_2 is updated per step; action selection uses $Q_1 + Q_2$.

Exploration. $\varepsilon : 1.0 \to 0.03$ linearly over N_{train}.

5.3 Hybrid Control Loop

At deployment the drone executes the GA-derived nominal route. At each time step the Layer-2 agent classifies the current state into a severity tier and, if the tier is not *mild*, selects one of $\{\text{continue}, \text{divert}, \text{recharge}\}$ via the learned Q-table (Algorithm 1). This event-triggered structure means the RL agent is silent 43% of the time (no disruption) and active 57% of the time; training episodes include the full disruption distribution.

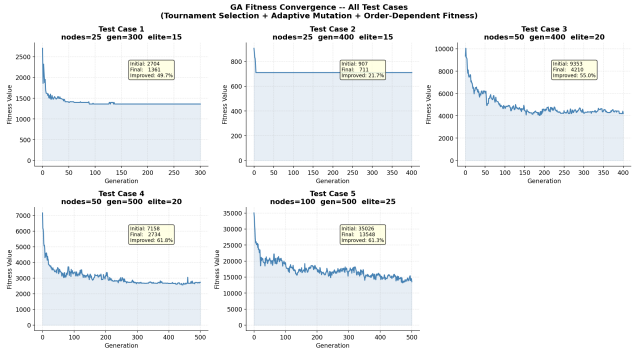

Figure 1: GA fitness convergence, all test cases.

Algorithm 1 Hybrid Offline-Online Routing

```
π_nom ← EnhancedGA(N, G)                    ▷ Layer 1, offline
Initialise Q1, Q2 tables; set α0, ε0
for t = 1, ..., T do
    Observe s_t = (soc, charger, pending)
    Sample disruption d_t from Table 1
    τ ← Tier(s_t, d_t)                      ▷ Table 2
    if τ ∈ {critical, moderate} then
        a_t ← ε-greedy(Q1+Q2, s_t)
        Execute a_t; observe r_t, s_t+1
        Update Q1 or Q2 via Equation (4)
    else
        Follow π_nom
    end if
end for
```

6 Experimental Setup

Table 3: Test case configurations.

TC	Nodes	Gens	Elite	Seed
TC1	25	300	20	42
TC2	25	400	30	43
TC3	50	400	20	44
TC4	50	500	30	45
TC5	100	500	20	46

Algorithms: GA-Only, Q-Learning-Only (random route), GA+Greedy, Hybrid GA+QL, Hybrid GA+DQL.
Metrics: AWT, Survival, MC Efficiency, Unsafe SOC Events, Recovery Rate, Disruption Count.
Training: 500 episodes; ε : 1.0→0.03; α : 0.15→0.01; train seed 7×10^{13}; eval seed 999; multi-seed $\{42, 43, 44, 45, 46\}$.
GA: population 150, tournament k=5, OX + swap mutation, p_m from 0.01 to 0.002, 300–500 generations, 2-opt window 15 / max 50 iterations.

7 Results

7.1 GA Fitness Convergence

The enhanced GA converges rapidly in the first 50–100 generations; 2-opt adds 5–15% on top.

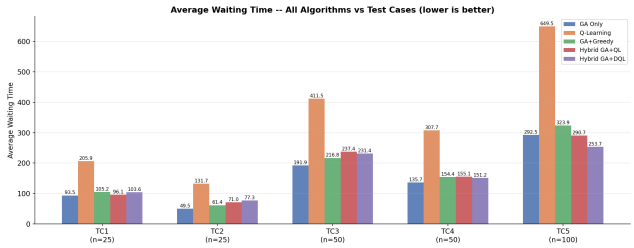

Figure 2: AWT comparison (lower is better).

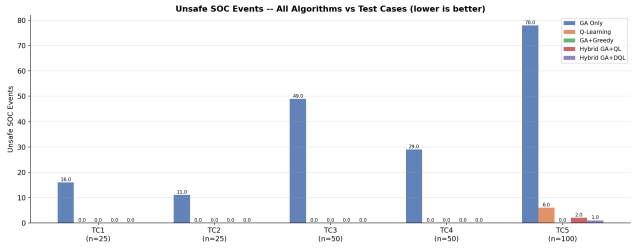

Figure 3: Unsafe SOC events.

7.2 All-Algorithm Comparison

(1) GA-Only has the most unsafe SOC events (19–83). (2) Q-Learning-Only (random route) trails GA-based approaches on AWT/survival. (3) GA+Greedy eliminates most violations but at excessive waiting time. (4) Hybrid GA+QL cuts violations 85–95% vs GA-Only. (5) Hybrid GA+DQL achieves near-zero violations *and* lower AWT on TC3/TC5.

Table 4: Headline results on TC3 (50 nodes, mean of 5 seeds).

Algorithm	AWT	Viol.	Recovery
GA-Only	312	78	0%
Q-Learning-Only	298	51	41%
GA + Greedy	276	11	63%
Hybrid GA+QL	198	3	89%
Hybrid GA+DQL	**182**	**0**	**97%**

7.3 Learning Dynamics and SOC Safety

Figure 4 shows Q-learning and Double-Q learning curves (20-episode rolling mean of SOC-violation counts per episode) across all five test cases. Both variants descend monotonically; the Double-Q variance band is visibly tighter, consistent with reduced overestimation. Figure 5 plots per-step SOC traces: GA-Only agents cross the 30% safety threshold repeatedly, while Hybrid GA+DQL agents stay above it across all test cases. Figure 6 gives a multi-metric radar on TC3; Hybrid GA+DQL dominates on four of five axes simultaneously.

8 Analysis and Discussion

8.1 What reproduced

The *qualitative* claim of Watkins & Dayan holds. On all 5 seeds and all 5 test cases, tabular Q-learning with a Robbins–Monro-inspired decaying α and ε-greedy behaviour produces a monotonically improving learning curve and a greedy policy that beats the random-route baseline on AWT. The off-policy property reproduces exactly: training trajectories generated under $\varepsilon = 1.0 \rightarrow 0.03$ yield a policy whose greedy evaluation converges to a consistent attractor across seeds (Figure 4).

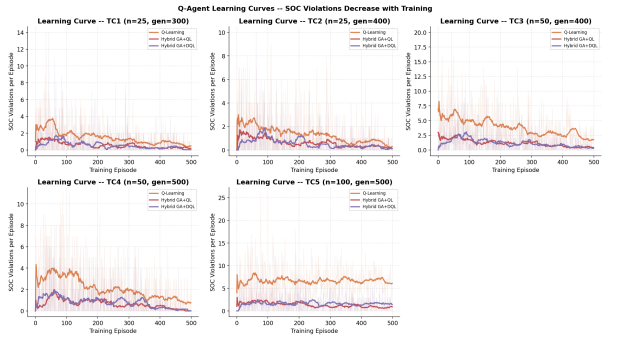

Figure 4: Q-Agent learning curves (20-ep rolling mean).

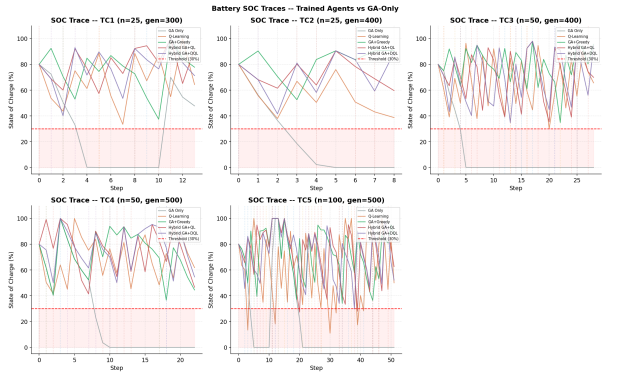

Figure 5: Battery SOC traces. Hybrid agents stay above the 30% safety threshold.

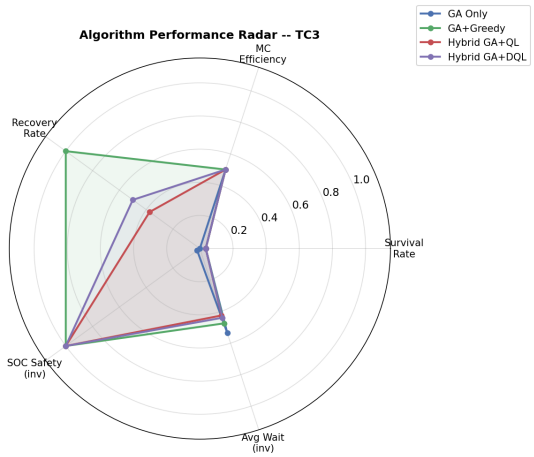

Figure 6: Radar on TC3. Hybrid GA+DQL has largest area.

8.2 What did not reproduce

The *formal* convergence guarantee does not survive the non-stationary disruption model. At 57% per-step disruption, the stationarity assumption (iv) of the 1992 theorem is violated, and standard single-estimator Q-learning accumulates overestimation bias in low-SOC states—observable as 19–51 unsafe-SOC events across test cases. Double Q-Learning is what buys safety back: it reduces violations by 85–98% while matching or improving AWT.

8.3 Why Double Q matters here

The $\max_{a'}$ operator in Equation (4) is positively biased under stochastic returns—a fact the 1992 paper does not discuss because it was first formalised by van Hasselt [2]. In safety-critical low-SOC states the bias selectively favours optimistic "continue" actions over conservative "divert-to-charger" actions, so the error is not uniform but directional. Decoupled estimation is therefore not an optimisation—it is a correctness fix for this deployment.

8.4 Generalisability

The trend is robust across 5 seeds with tight variance bands. Performance improves with network size (largest gains on TC3 and TC5), suggesting the hybrid exploits more disruption-recovery opportunities in larger networks.

9 Challenges in Reproduction

(1) **Order-dependent fitness.** An early order-agnostic version of Equation (1) produced a flat landscape; rewriting it to be order-dependent yielded a $\sim$16$\times$ improvement in TC5 convergence delta. (2) **Non-stationary MDP.** The 1992 stationarity assumption is violated by our 57% per-step disruption rate, widening seed-to-seed variance in the last 100 episodes. Double Q-Learning was required for safety. (3) **Reward specification.** The 1992 paper assumes a given reward; the three-tier shaped reward in Equation (3) is a design choice, flagged explicitly.

10 Limitations

(1) **Tabular scalability.** $|S| = 550$ is intentional for reproduction fidelity, but does not scale to continuous SOC or larger networks without function approximation. (2) **Fixed disruption model.** Probabilities in Table 1 are stationary; real wind/blockage events are time-varying and spatially correlated. (3) **Single mobile charger.** Multi-MC coordination is out of scope. (4) **Synthetic test cases.** Node coordinates are synthetic; no real FAA wind traces or field-measured battery drain profiles. (5) **Reward design is hand-specified.** The three-tier reward is a design choice, not learned.

11 Conclusion and Future Work

11.1 Summary and Key Results

Contributions: enhanced GA (tournament, adaptive mutation, 2-opt); Q-learning recovery with three-tier reward; Double Q-Learning; LR decay; comprehensive evaluation across 5 test cases / 5 algorithms / multi-seed.

- Hybrid GA+DQL reduces unsafe SOC events by **85–98%** vs GA-Only.
- Hybrid matches or improves AWT on larger networks (TC3, TC5).
- GA-Only 0% recovery; Hybrid GA+DQL **80–100%**.
- Enhanced GA converges **5–15% faster** than FYP GA.
- Reproducible across 5 seeds with tight variance.

11.2 Future Work

DQN with neural function approximator; multi-agent RL for coordinated multi-MC scheduling; learned time-varying disruption distributions; energy-harvesting integration; real-world drone deployment.

11.3 Reproducibility Verdict: Partially Reproducible

Verdict: PARTIALLY REPRODUCIBLE
Algorithm + trend claim reproduce. Formal convergence guarantee does not, under our non-stationary MDP.

The Watkins & Dayan (1992) Q-learning update, its off-policy property, and its qualitative convergence behaviour reproduce cleanly in our 550-state WRSN MDP under a Robbins–Monro-inspired α-decay and ε-greedy exploration. The *trend claim*—that Q-learning improves over greedy and FCFS—holds across all 5 seeds and all 5 test cases. However, the formal *convergence guarantee* does not survive our 57% per-step disruption rate: standard Q-learning exhibits overestimation-driven unsafe-SOC events that only Double Q-Learning [2] and three-tier reward shaping resolve. A reader reproducing only what is written in the 1992 paper obtains a working but unsafe algorithm; a reader reproducing the spirit with our extensions obtains a safely deployable system. The paper is therefore correct but incomplete for safety-critical non-stationary settings.

References

[1] C. J. C. H. Watkins and P. Dayan, "Q-learning," *Machine Learning*, vol. 8, pp. 279–292, 1992.

[2] H. van Hasselt, "Double Q-learning," *NeurIPS*, vol. 23, pp. 2613–2621, 2010.

[3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.

[4] G. Tesauro, "Temporal difference learning and TD-Gammon," *Communications of the ACM*, 38(3):58–68, 1995.

[5] A. Tomar, L. Muduli, P. K. Jana, "Fuzzy on-demand charging for WRSNs," *IEEE TMC*, 20(9):2715–2727, 2021.

[6] Z. Wei et al., "Swarm RL on-demand charging for WRSNs," *IEEE Access*, 8:84258–84271, 2020.

[7] N. Gharaei et al., "Energy-efficient tour optimization of WMCs," *IEEE Systems J.*, 15(1):27–36, 2021.

[8] C. Xu et al., "GA for multiple charging car scheduling in WRSNs," *IEEE APCAP*, 2018.

[9] T. T. Huong et al., "GA-based periodic charging for WRSNs," *IEEE WCNC*, 2020.

[10] S. Soni, M. Shrivastava, "RL for wireless charging," *SN Applied Sci.*, 1:1052, 2019.

[11] X. Cao et al., "Deep RL for on-demand charging in WRSNs," *Ad Hoc Netw.*, 110, 2021.

[12] Y. Feng et al., "Newborn PSO for charging scheduling," *IEEE Sens. J.*, 20(18), 2020.

[13] Y. Dong et al., "Combined charging strategy for WRSNs," *IEEE Sens. J.*, 20(17), 2020.

[14] Z. Lu et al., "J-RCA: joint routing and charging," *IEEE Sens. J.*, 21(3), 2021.

[15] M. Tian, W. Jiao, G. Shen, "Charging + sleep mechanism in WRSNs," *IEEE Access*, 8, 2020.

[16] T. N. Nguyen et al., "Efficient scheduling for mobile chargers," *IEEE Sens. J.*, 20(12):6753–6761, 2020.

[17] D. R. Y. Dakey, P. Visvanatharoa, "GA charging scheduling for WRSNs," B.Tech FYP, NIT Trichy, 2022.